

Literature-implied partial answer for arXiv:2206.10279

The skein free space is not Dirac-witnessed Plichko

Status. `literature_implied_answer` (partial subcase).

Source question. The source is P. Hajek and A. Quilis, *A complete metric space without non-trivial separable Lipschitz retracts*, arXiv:2206.10279 [1]. In the Introduction, parsed source lines 110–115, the authors recall the Plichko property and the Separable Complementation Property (SCP), state that they do not know examples of Lipschitz-free spaces over metric spaces failing to be Plichko, and ask in particular whether the Lipschitz-free space over their constructed metric space is Plichko, or even has the SCP.

Supporting theorem. The supporting paper is A. J. Guirao, V. Montesinos, and A. Quilis, *On projectional skeletons and the Plichko property in Lipschitz-free Banach spaces*, arXiv:2305.08543 [2]. Theorem 2.1 of [2] says, among other equivalent conditions, that for a complete pointed metric space M and $\lambda \geq 1$, the free space $\mathcal{F}(M)$ is λ -Plichko witnessed by a subset of $\delta(M)$ if and only if for every $p \in M$ and every $r < 1/\lambda$, the ball

$$B(p, r d(p, 0))$$

is separable.

Claim. Let $M = \text{Sk}(\omega_1)$ be the skein metric space constructed in [1], pointed at the initial anchor A . Then $\mathcal{F}(M)$ is not λ -Plichko witnessed by Dirac measures for any finite $\lambda \geq 1$.

Proof. Let B be the other initial anchor of the 0-skein. In the construction of [1], $d(A, B) = 1/2$, and the first successor stage attaches the threading space $\text{Th}(A, B)$. This threading space contains, for every $\gamma \in \Delta$, a thread $T_\gamma(A, B)$ of width $1/2$; different such threads meet exactly in the two anchors A, B . The set Δ is uncountable.

We first note that B is not isolated inside any $T_\gamma(A, B)$. Indeed, in the real model $T_\gamma \subset [0, 1]$, the complement is the union of the gaps $G_i^\gamma = (q_{n_i}, q_{n_i} + \gamma_i)$, each lying strictly inside $(0, 1)$, and Proposition 2.6 of [1] identifies these gaps as the components of the complement. If a terminal interval $(1 - \varepsilon, 1)$ were disjoint from T_γ , connectedness of that interval would force it to lie in a single complementary component whose right endpoint is 1, which is impossible.

Fix $\lambda \geq 1$. Choose $r > 0$ with

$$r < \min\{1/\lambda, 1/4\},$$

and put $\rho = r d(B, A) = r/2$. For each $\gamma \in \Delta$, choose a non-anchor point $x_\gamma \in T_\gamma(A, B)$ such that

$$0 < d(x_\gamma, B) < \rho.$$

Split these choices into the countably many annuli

$$\frac{\rho}{n+1} \leq d(x_\gamma, B) < \frac{\rho}{n}, \quad n = 1, 2, \dots$$

Since Δ is uncountable, one annulus contains x_γ 's for an uncountable set $\Delta_0 \subset \Delta$. If $\gamma \neq \eta$ are in Δ_0 , the attachment formula for distinct threads in $\text{Th}(A, B)$ gives

$$d(x_\gamma, x_\eta) = \min\{d(x_\gamma, A) + d(A, x_\eta), d(x_\gamma, B) + d(B, x_\eta)\}.$$

The second term is at least $2\rho/(n+1)$. The first term is bounded below by $2(1/2 - \rho) > 0$, because $d(A, B) = 1/2$ and $d(x_\gamma, B) < \rho < 1/8$. Thus the set $\{x_\gamma : \gamma \in \Delta_0\}$ is uncountable and uniformly separated, while it is contained in $B(B, \rho) = B(B, r d(B, A))$. Therefore this ball is nonseparable.

The local separability condition in Theorem 2.1 of [2] fails for the point $p = B$ and the above $r < 1/\lambda$. Hence $\mathcal{F}(M)$ is not λ -Plichko witnessed by a subset of $\delta(M)$. Since $\lambda \geq 1$ was arbitrary, no finite Dirac-witnessed Plichko constant is possible. $\square$

Scope and limitations. This is not a full answer to the source question. It rules out only Plichko witnesses supported on Dirac measures. It does not rule out a Plichko witness using molecules or more general vectors in $\mathcal{F}(M)$, and it does not decide whether $\mathcal{F}(M)$ has the SCP.

Search and provenance. The relation above is not stated as an answer to arXiv:2206.10279 in [2]; it is an agent-identified implication of Theorem 2.1 combined with the local geometry of the skein construction. Web searches for “Lipschitz-free spaces fail to be Plichko”, “F(ell.infty) Plichko SCP”, and close variants found the 2023 supporting paper, which still states that the unrestricted Plichko and SCP questions remain open.

References

- [1] P. Hajek and A. Quilis, *A complete metric space without non-trivial separable Lipschitz retracts*, arXiv:2206.10279, 2022.
- [2] A. J. Guirao, V. Montesinos, and A. Quilis, *On projectional skeletons and the Plichko property in Lipschitz-free Banach spaces*, arXiv:2305.08543, 2023.